

## Test 12 — Question Paper

**38 items.** Suggested time: 95 minutes (about 2 minutes 30 seconds per item).

A four-function calculator may be used **only** on items marked *calculator*.

The Grade 6 B.E.S.T. Mathematics Reference Sheet is on the last page.

Aligned to the FAST Grade 6 Mathematics test design: 33–42% Number Sense and Operations, 25–36% Algebraic Reasoning, 25–36% Geometric Reasoning, Data Analysis, and Probability.

1. Elena had some stickers and gave away 9 of them. Now Elena has 11 stickers.

Which equation can be used to find  $x$ , the number of stickers Elena started with?

- A.  $9 - x = 11$
- B.  $x + 11 = 9$
- C.  $x - 9 = 11$
- D.  $x + 9 = 11$

Show your work

multiple choice

2. Point A is located at  $(-5, -5)$ . Point A is reflected across the  $x$ -axis.

What are the coordinates of the reflected point?

- A.  $(-5, 5)$
- B.  $(5, 5)$
- C.  $(-3, 5)$
- D.  $(-5, -5)$

Show your work

multiple choice

3. Which expression is equivalent to  $9x + 4 + 8x$ ?

- A.  $72x + 4$
- B.  $17x + 4$
- C.  $17x + 4x$
- D.  $21x$

Show your work

multiple choice

4. A triangle has an area of 25 square feet and a height of 5 feet.

What is the length of its base, in feet?

- A. 12
- B. 10
- C. 5
- D. 50

Show your work

multiple choice

5. Gabriel recorded these four numbers in her math journal: -3.5,  $-1\frac{1}{4}$ ,  $\frac{1}{8}$ , 3.

Which number is the least?

- A. -3.5
- B.  $-1\frac{1}{4}$
- C.  $\frac{1}{8}$
- D. 3

Show your work

calculator · multiple choice

6. What value of  $x$  makes the equation  $\frac{5}{6} \times x = \frac{5}{4}$  true?

- A.  $\frac{3}{2}$
- B.  $\frac{25}{12}$
- C.  $\frac{5}{9}$
- D.  $\frac{25}{24}$

Show your work

multiple choice

7. A data set is shown.

1, 3, 4, 5, 5, 7, 9, 10, 12, 13, 13, 14

A histogram is made with the intervals 0-4, 5-9, 10-14 and 15-19.

How tall is the bar for the interval 15-19?

- A. 12
- B. 0
- C. 1
- D. 2

Show your work

multiple choice

8. The table shows equivalent part-to-part-to-whole ratios.

What number belongs in the box marked ?

| Boys | Girls | Total |
|------|-------|-------|
| 4    | 7     | 11    |
| 12   | 21    | ?     |

Part-to-part-to-whole ratios

- A. 19
- B. 36
- C. 33
- D. 11

Show your work

multiple choice

9. Which value of x makes the inequality  $x > 27$  true?

- A. 28
- B. 21
- C. 25
- D. 27

Show your work

multiple choice

10. What is the value of  $|-20 \frac{3}{10}|$ ?

- A. -0.05
- B. 20.3
- C. -20.3
- D. 40.6

Show your work

calculator · multiple choice

11. A bag holds 21 mangoes and 12 guavas.

What is the ratio of mangoes to the total number of pieces of fruit, in simplest form?

- A. 7:11
- B. 7:4
- C. 4:7
- D. 11:7

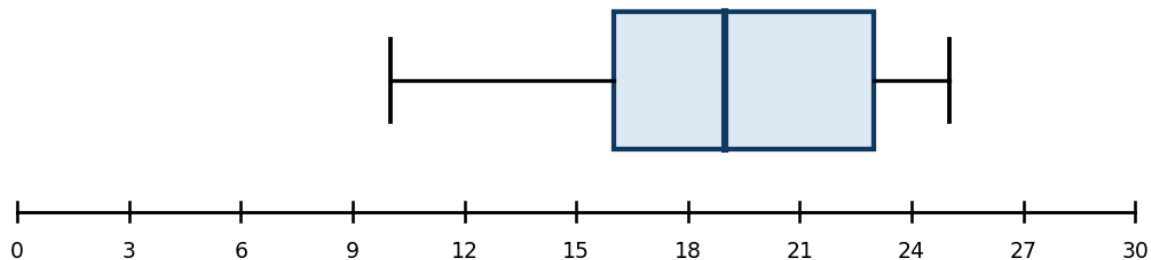
Show your work

multiple choice

12. The box plot shows a set of data.

What is the range of the data?

**Books read**



- A. 23
- B. 19
- C. 15
- D. 7

Show your work

multiple choice

13. What is the value of  $2^3$ ?

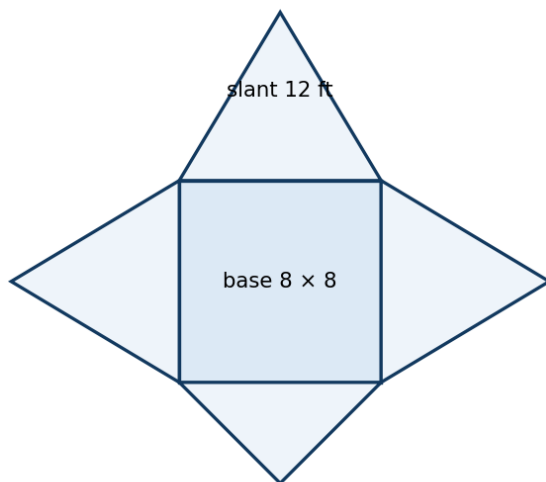
- A. 4
- B. 11
- C. 6
- D. 8

Show your work

calculator · multiple choice

14. The net of a right rectangular pyramid with a square base is shown.

What is the surface area of the pyramid, in square feet?



- A. 64
- B. 384
- C. 448
- D. 256

Show your work

multiple choice

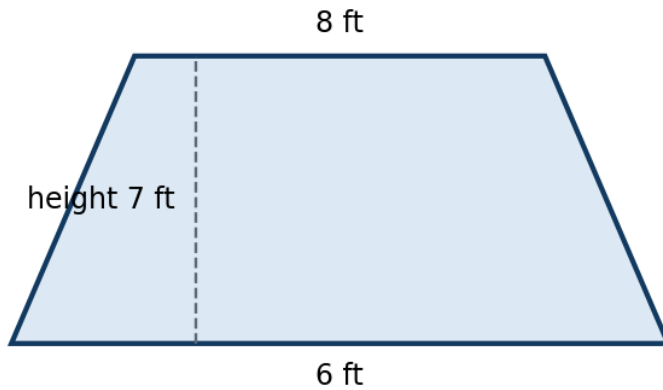
15. Which expression shows the prime factorization of 252?

- A.  $2 \times 127$
- B.  $2^2 \times 3^2 \times 7$
- C.  $2 \times 3^2 \times 7$
- D.  $2^3 \times 3^2 \times 7$

Show your work

calculator · multiple choice

16. What is the area of the trapezoid shown, in square feet?



- A. 42
- B. 98
- C.  $10 \frac{1}{2}$
- D. 49

Show your work

multiple choice

17. Ana buys 9 cases of water at \$4.45 each and pays a \$3.47 delivery fee.

What is the total cost?

- A. \$71.28
- B. \$35.68
- C. \$43.52
- D. \$40.05

Show your work

calculator · multiple choice

18. Select **all** the expressions with a value of -29.

- ☐  $-17 - (-12)$
- ☐  $(-12) + -17$
- ☐  $17 + (-12)$
- ☐  $-17 \times (-12)$
- ☐  $-17 - (12)$
- ☐  $-17 + (-12)$

Select all that apply.

Show your work

calculator · multiselect

19. Which question is **not** a statistical question?

- A. How many text messages did each student send yesterday?
- B. How tall is the tallest student in my class?
- C. How many hours did each student in the class sleep last night?
- D. How many minutes did each sixth grader spend on the bus this morning?

Show your work

multiple choice

20. What is the product of  $\frac{3}{5}$  and  $\frac{3}{8}$ ?

- A.  $\frac{6}{13}$
- B.  $\frac{39}{40}$
- C.  $\frac{9}{40}$
- D.  $1\frac{3}{5}$

Show your work

calculator · multiple choice

21. A box shaped like a right rectangular prism has a volume of 140 cubic inches. Its base has an area of 70 square inches.

What is the height of the box, in inches?

- A. 2
- B. 4
- C. 70
- D. 0

Show your work

multiple choice

22. A pitcher holds 8 cups of juice.

Select **all** the expressions that give the number of fluid ounces the pitcher holds. (1 cup = 8 fluid ounces)

- ☐  $8 \times 2$
- ☐  $8 \times 8$
- ☐  $8 \div 8$
- ☐  $8 + 8$
- ☐  $64 \times 1$

Select all that apply.

Show your work

multiselect

23. What is the greatest common factor of 45 and 75?

- A. 225
- B. 15
- C. 30
- D. 45

Show your work

calculator · multiple choice

24. What is the value of  $(3x + 49) \div 5$  when  $x = 2$ ?

- A. 50
- B. 11
- C. 15
- D. 55

Show your work

multiple choice

25. Select **all** the expressions with a value of 48.

- ☐  $(6) \times (8)$
- ☐  $(-6) \div (-8)$
- ☐  $(-6) \times (-8)$
- ☐  $(-6) + (-8)$
- ☐  $(-8) \times (-6)$
- ☐  $(6) \times (-8)$

Select all that apply.

Show your work

calculator · multiselect



**26.** A diver off Crandon Beach descends 17 feet below the surface, then rises 11 feet. Sea level is represented by 0.

Which integer represents the diver's position now?

- A. -6
- B. 28
- C. -28
- D. 6

Show your work

calculator · multiple choice

**27.** Select **all** the forms that are equivalent to  $\frac{5}{8}$ .

- ☐ 6.25
- ☐  $\frac{8}{5}$
- ☐  $\frac{5}{8}$
- ☐ 0.625
- ☐ 625%
- ☐ 62.5%

Select all that apply.

Show your work

calculator · multiselect

**28.** Lucia wrote the expression  $x - 2$ .

Which written description matches the expression?

- A. the product of a number  $x$  and 2
- B. 2 less than a number  $x$
- C. a number  $x$  decreased by 2
- D. a number  $x$  divided by 2

Show your work

multiple choice

**29.** What is the product of 15.2 and 0.69?

- A. 1.0488
- B. 1048.8
- C. 104.88
- D. 10.488

Show your work

calculator · multiple choice

**30.** Which percent is equivalent to  $\frac{3}{20}$ ?

- A. 15%
- B. 666.6667%
- C. 1.5%
- D. 25%

Show your work

calculator · multiple choice

**31.** Javier recorded these scores:

6, 7, 8, 8, 8, 14, 18

What is the median of the data set?

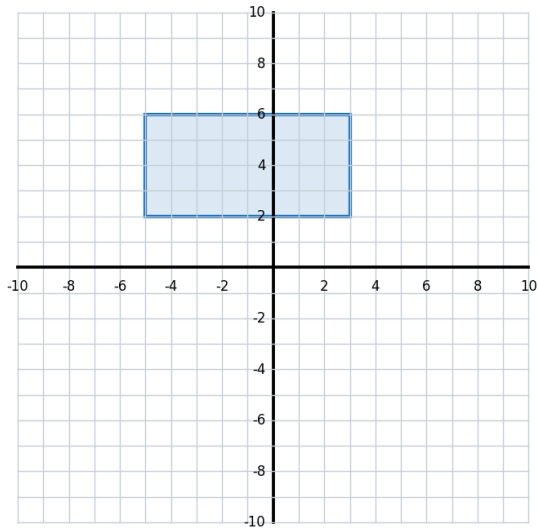
- A. 12
- B. 8
- C. 9
- D. 10

Show your work

multiple choice

**32.** A rectangle has vertices at  $(-5, 2)$ ,  $(3, 2)$ ,  $(3, 6)$  and  $(-5, 6)$ .

What is the perimeter, in units, of the rectangle?



- A. 32
- B. 24
- C. 12
- D. 64

Show your work

multiple choice

**33.** Paola shared 48 guavas equally among 4 baskets.

Which equation can be used to find  $x$ , the number of guavas in each basket?

- A.  $x + 4 = 48$
- B.  $48x = 4$
- C.  $4x = 48$
- D.  $x \div 4 = 48$

Show your work

multiple choice

**34.** Four sensors on a reef near Key Biscayne report these elevations, in feet relative to sea level:

-63, -58, -34, -25

Which elevation is farthest from sea level?

- A. -58
- B. -25
- C. -63
- D. -34

Show your work

calculator · multiple choice

**35.** The number line shows the solution to an inequality.

Which inequality does the graph represent?



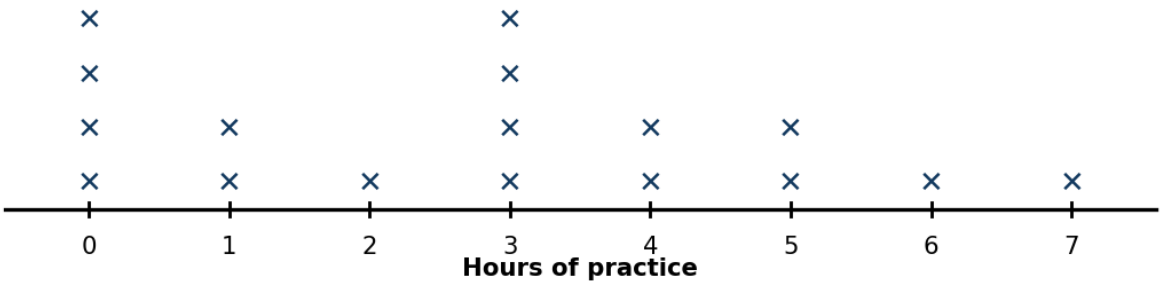
- A.  $x \leq 36$
- B.  $x < 36$
- C.  $x \geq 36$
- D.  $x > 36$

Show your work

multiple choice

36. The line plot shows hours of practice.

What is the range of the data?



- A. 9
- B. 17
- C. 7
- D. 8

Show your work

multiple choice

37. A bus travels 180 miles in 2 hours at a constant speed.

At this rate, how far does the bus travel in 1 hour?

- A. 1/90 miles
- B. 360 miles
- C. 90 miles
- D. 178 miles

Show your work

multiple choice

**38.** Rewrite  $28 + 44$  as the greatest common factor of the two numbers multiplied by the sum of two whole numbers.

Which expression is correct?

- A.  $4(7 + 12)$
- B.  $4(18)$
- C.  $7(4 + 11)$
- D.  $4(7 + 11)$

Show your work

calculator · multiple choice

## Grade 6 Mathematics Reference Sheet

### Customary Conversions

|                      |                        |
|----------------------|------------------------|
| 1 foot = 12 inches   | 1 cup = 8 fluid ounces |
| 1 yard = 3 feet      | 1 pint = 2 cups        |
| 1 mile = 5,280 feet  | 1 quart = 2 pints      |
| 1 mile = 1,760 yards | 1 gallon = 4 quarts    |
| 1 pound = 16 ounces  | 1 ton = 2,000 pounds   |

### Metric Conversions

|                             |                             |
|-----------------------------|-----------------------------|
| 1 meter = 100 centimeters   | 1 liter = 1,000 milliliters |
| 1 meter = 1,000 millimeters | 1 gram = 1,000 milligrams   |
| 1 kilometer = 1,000 meters  | 1 kilogram = 1,000 grams    |

### Time Conversions

|                       |                   |
|-----------------------|-------------------|
| 1 minute = 60 seconds | 1 week = 7 days   |
| 1 hour = 60 minutes   | 1 year = 365 days |
| 1 day = 24 hours      | 1 year = 52 weeks |

### Formula

|                                         |                                  |
|-----------------------------------------|----------------------------------|
| Rectangular prism                       | $V = lwh$ or $V = Bh$            |
| $l$ = length, $w$ = width, $h$ = height | $B$ = area of base, $V$ = volume |